Arithmetic Applicability of Classical Odd-Run Merges

Collatz proof project — September 5, 2026

This note formalizes the classical consecutive-seed endpoint identities and applies them to the current proof program. The merging stem is due to Garner; the explicit endpoint dichotomy appears in LaDue's Lemma 3.1 and Theorem 4.1 [1]. No mathematical novelty or global coverage is claimed. Throughout, $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n.

Theorem 1 (Odd-run endpoint dichotomy). *Let $a \geq 1$, let m be a positive odd integer, and put $n = 2^a m - 1$. The first a steps from n are odd, and*

$$T^a(n) + 1 = 3^a m, \qquad T^a(n) = 3T^a(n-1) + 2.$$

At time $a+2$, exactly one of the following holds:

$3^a m \pmod 4$	*endpoint relation*
1	$T^{a+2}(n) = T^{a+2}(n-1)$,
3	$T^{a+2}(n) = 9T^{a+2}(n-1) + 2$.

In particular, equality at that time holds if and only if $3^a m \equiv 1 \pmod 4$.

Proof. For an odd input, $T(n)+1 = 3(n+1)/2$. Repeating this identity proves the odd-run formula. The seed $n-1$ first takes an even step, followed by $a-1$ odd steps, giving $T^a(n-1)+1 = 3^{a-1}m$. This proves the ratio-three identity. Both resulting endpoints are even. If $3^a m$ is one modulo four, the path from n takes two even steps and the path from $n-1$ takes an even and an odd step; the endpoints agree. If $3^a m$ is three modulo four, these two patterns are reversed, and direct substitution gives the ratio-nine identity. The latter endpoints cannot agree for natural seeds.

In Lean, `realizes_odd_run` proves actual parity realization, not just a formal affine formula. The arithmetic merge lemma applies the existing stem theorem. The remaining branch and the equivalence are proved separately by exact natural arithmetic. □

Theorem 2 (Necessary exit for a least unbounded seed). *Suppose $n > 1$ is the least positive seed with an unbounded orbit, and $n + 1 = 2^a m$ with $a \geq 1$ and m odd. Then $3^a m \equiv 3 \pmod 4$. Thus its initial odd run must be followed by a single even step and then an odd step, rather than two even steps.*

Proof. In the double-even case, the first theorem merges n with $n-1$. Minimality makes the positive seed $n-1$ bounded, so their common tail is bounded, contradicting unboundedness of n. No noncontraction property is needed for the replacement. This distinction matters: $n-1$ is even and its first multiplicative coefficient is $1/2$. The argument concerns the least unbounded seed, not the least seed whose coefficients never contract. □

Breaking a finite-table limitation. The previous fixed-table obstruction uses seeds with $n \equiv -1 \pmod{2^{K+18}}$ and $n \equiv 0 \pmod{3^{11}}$. For any $a \geq K + 18$, choose the odd part m to satisfy

$$m \equiv 2^{-a} \pmod{3^{11}}, \qquad m \equiv 3^{-a} \pmod 4.$$

CRT gives infinitely many such m. The resulting seeds remain in the table's obstruction through horizon K, yet the classical variable-length stem merges them with $n-1$ after $a+2$ steps. Choosing instead $m \equiv 3 \cdot 3^{-a} \pmod 4$ gives the single-even branch within the same obstruction. The generic endpoint theorems are kernel checked; this CRT application is a written deduction with exact Python instance checks.

What remains open. For a least unbounded n, the remaining branch gives $9z + 2$ on its orbit and z on the bounded orbit of $n-1$. Nothing here proves that boundedness of z implies boundedness of $9z+2$. That missing implication, or a different argument covering the branch, is substantive. Neither nondivergence nor exclusion of nontrivial cycles follows from these results.

Verification and source. Seven declarations in `Collatz.OddRunMerges` enter the axiom audit.
`analysis/odd_run_merges.py` constructs both branches, with tests of actual orbits and table applicability. These are instance checks, not a finite census claimed to prove a universal statement.

[1] M. D. LaDue, *Clusters of Integers with Equal Total Stopping Times in the $3x+1$ Problem* (2017), Lemma 3.1 and Theorem 4.1. arXiv:1709.02979.